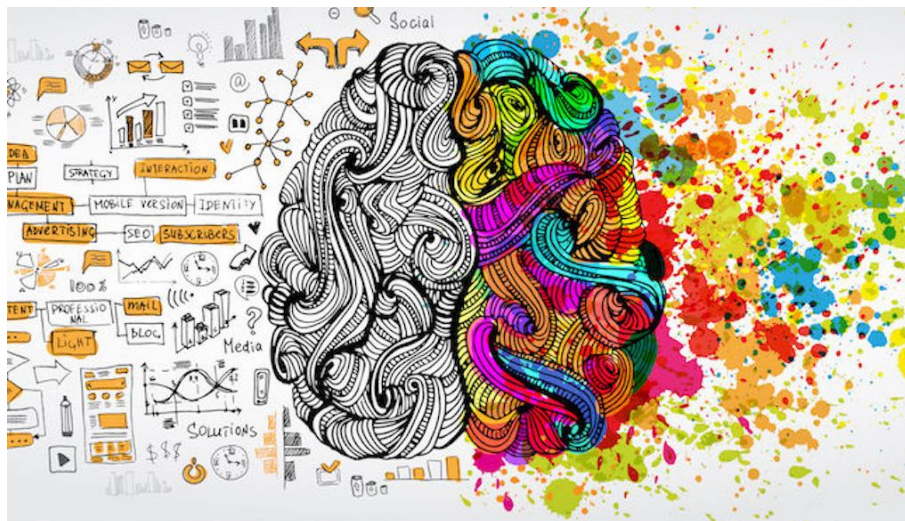




Legal Design and Digital Rights in AI

Monica Palmirani, University of Bologna,
CIRSFID-AlmaAI
Hackathon 2026

Four main outcomes to take away

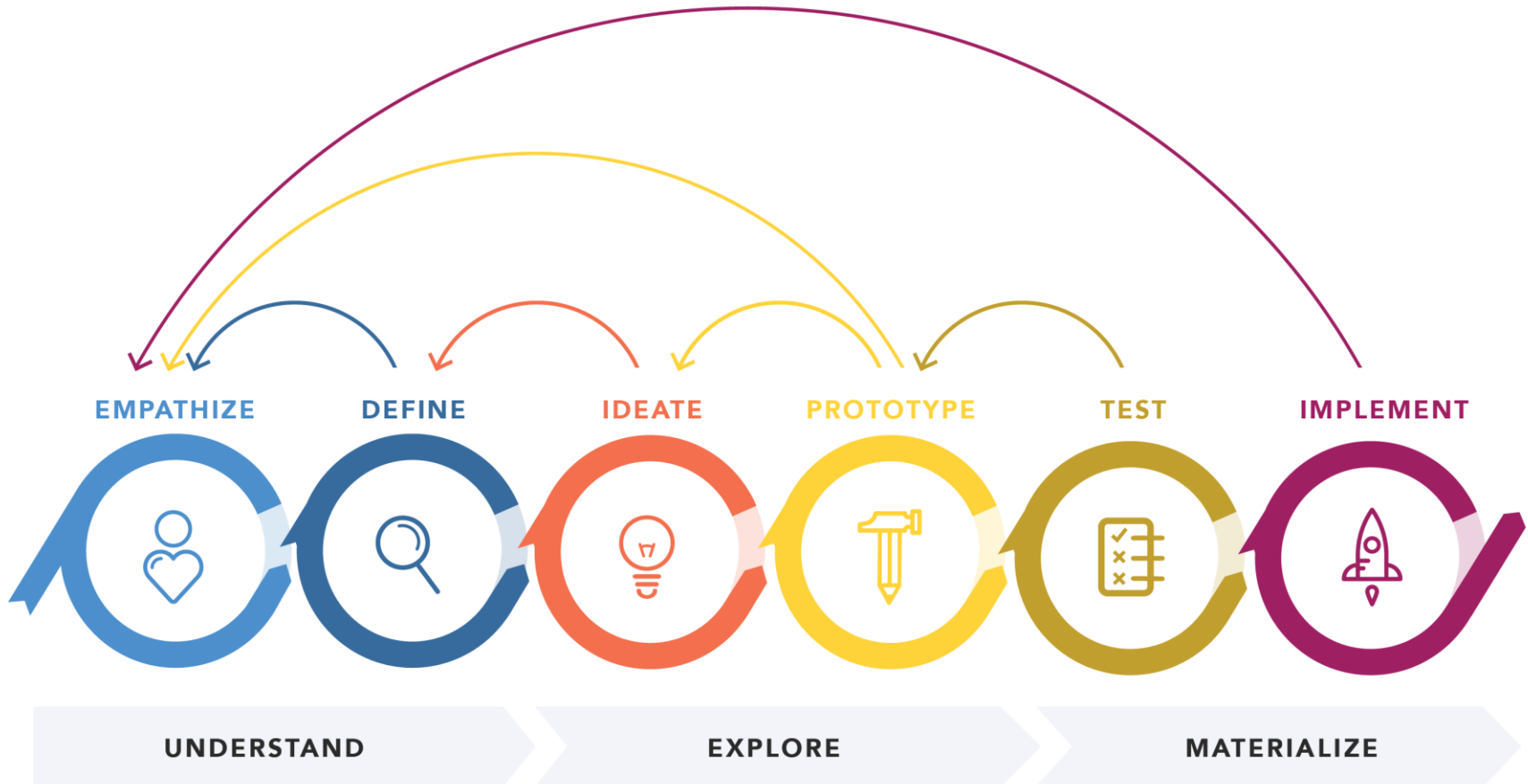
#1 Normativity with non-textual language

#2 Transparency and Explicability: Legal ontology

#3 Digital Right: Chapter III “Interactions with algorithms and artificial intelligence systems”

#1 NORMATIVITY WITH NON-TEXTUAL LANGUAGE

Design Thinking



Normativity

- Normativity is the **power** of the **authority** to release command addressed to citizens
- The command should be **transparent** and **understandable**
- Explain to the people the correct information concerning the norms and which **behaviour** is correct and which is not correct
- **Interior** Normativity and exterior Normativity (Hart 1969).
Digest the normativity and to transform in **behaviour**
- **Autonomy** to decide and to disobey, the disobedience can create new view and norms

Textual legal language

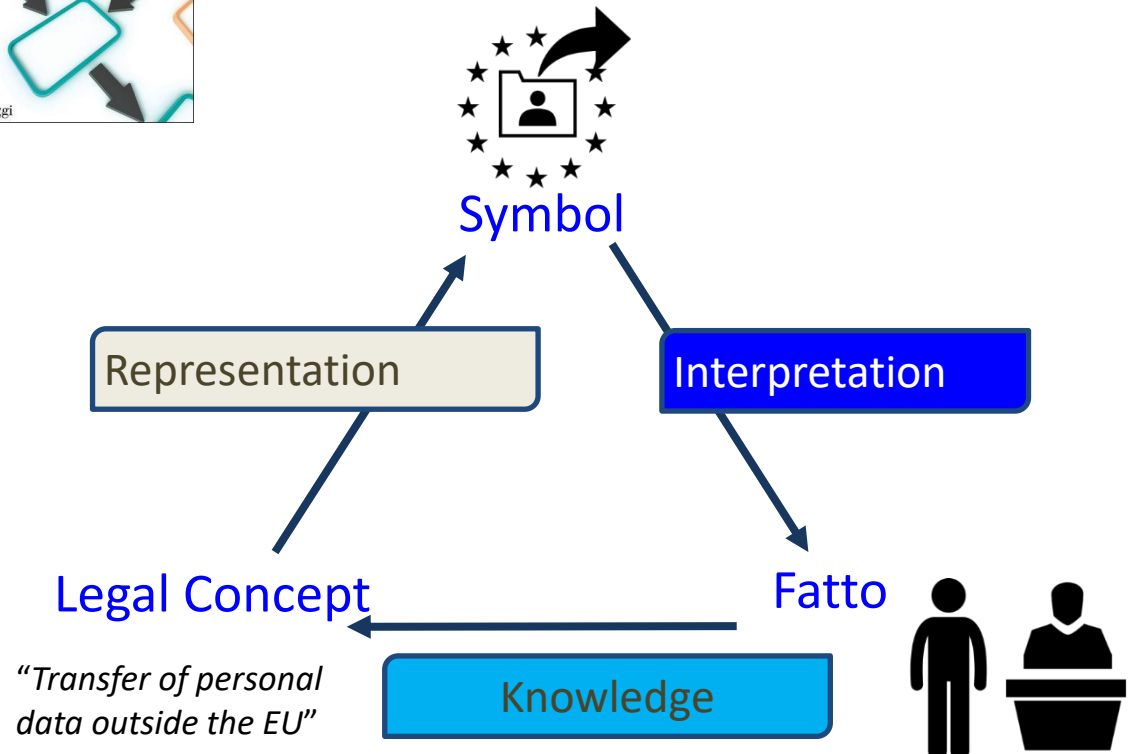
- For centuries textual legal language was the official legal instruments for creating normativity
- Good practices were developed using text legal language: thesauri, definitions, derogations, exceptions, modifications, etc.
- **Is non-textual language normative? And under which conditions?**

Peirce semiotic triangle



Antiphon of
Ramnunte, 4th
century BC, "If the
language changes,
the thought
changes"

Antoine Garapon



Visual norms in context



shutterstock.com · 1998974654



Metaphor

METAFORA

meta = out e *phèrein* = carry

The visual metaphor shifts the complex concept to another simplified concept

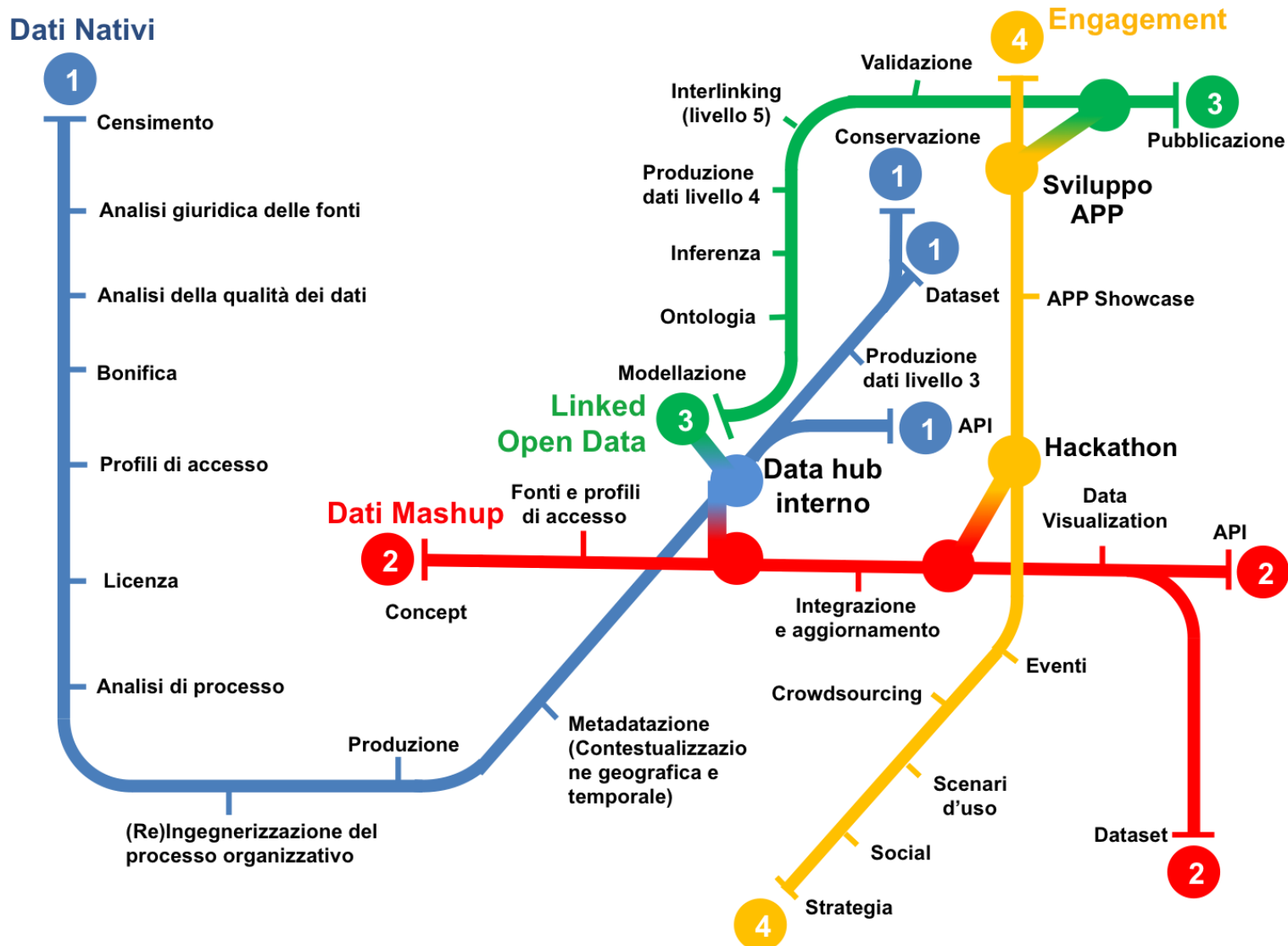
Condensation – Shifting - Association

<http://www.vismet.org/VisMet/>

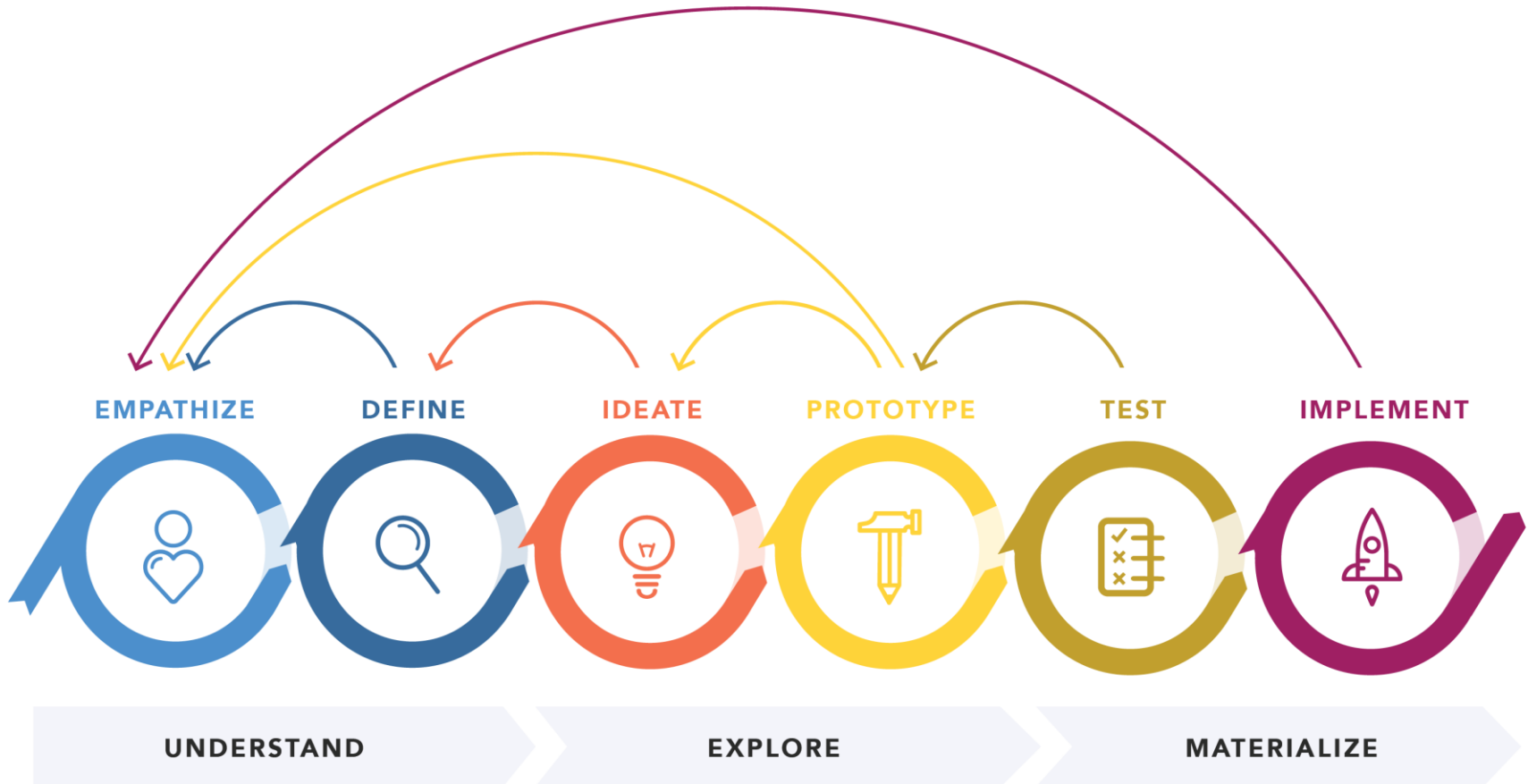
«Legal Person»



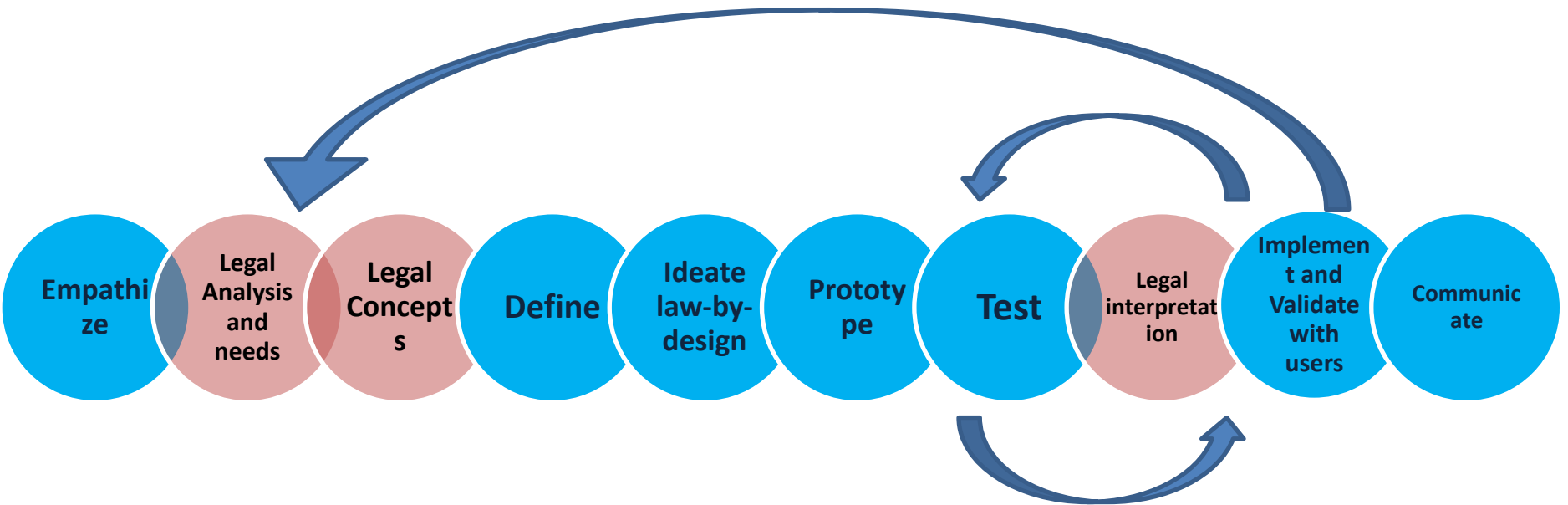
Metro Metaphor



Design Thinking



Legal Design thinking



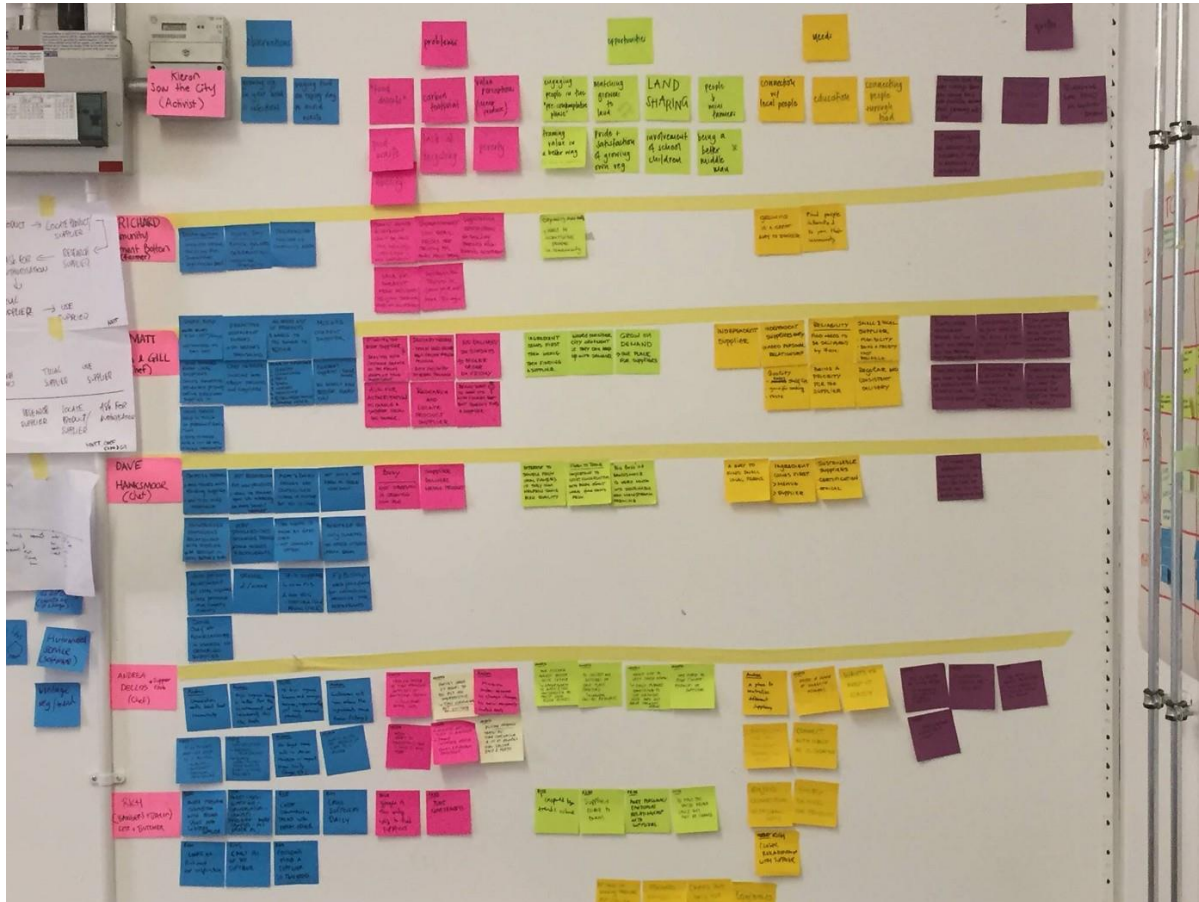
Legal Design: a possible methodology

- Problem analysis
 - Legal text/message analysis
 - Hypothesis of solution
 - Legal **concepts** extraction ****
 - » **Theory of law check** ***
 - » **Legal Ontology analysis** ***
 - Simplification of the scenario
 - Usability (**HCI**)
 - Visualization/Solution (**Design**)
 - Communication (**Language**)
 - Testing
 - Interpretation of the results
- Evaluation of the effect on the problem

SEMANTIC



Wall of post-it



#2 TRANSPARENCY AND EXPLICABILITY: LEGAL ONTOLOGY



THE EUROPEAN DATA STRATEGY

SHAPING EUROPE'S
DIGITAL FUTURE



SUSTAINABILITY

Digital devices should support sustainability and the green transition. People need to know about the environmental impact and energy consumption of their devices.



SAFETY AND SECURITY

The digital environment should be safe and secure. All users, from childhood to old age, should be empowered and protected.



PEOPLE AT THE CENTRE

Digital technologies should protect people's rights, support democracy, and ensure that all digital players act responsibly and safely. The EU promotes these values across the world.



SOLIDARITY AND INCLUSION

Technology should unite, not divide, people. Everyone should have access to the internet, to digital skills, to digital public services, and to fair working conditions.



FREEDOM OF CHOICE

People should benefit from a fair online environment, be safe from illegal and harmful content, and be empowered when they interact with new and evolving technologies like artificial intelligence.



PARTICIPATION

Citizens should be able to engage in the democratic process at all levels, and have control over their own data.

DIGITAL RIGHTS AND PRINCIPLES

Chapter III: Freedom of choice

Interactions with algorithms and artificial intelligence systems

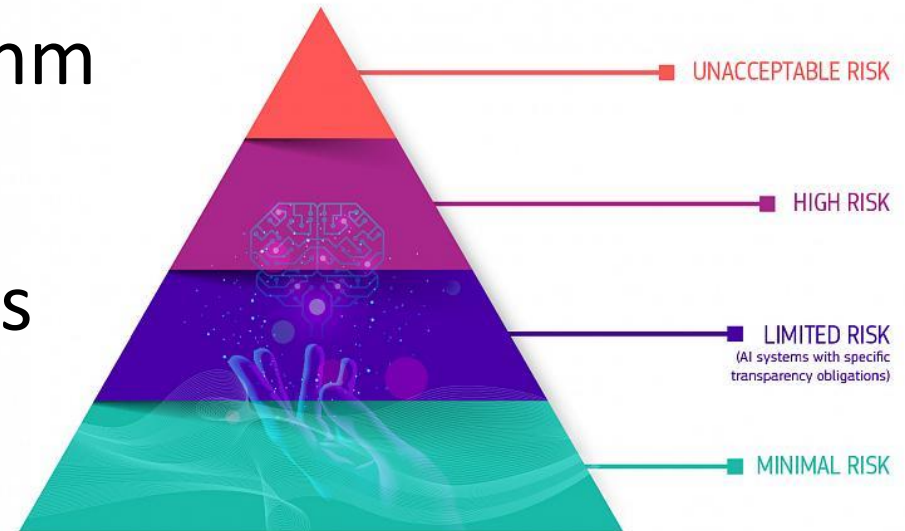
8. Artificial intelligence should serve as a tool for people, with the ultimate aim of increasing human well-being.
9. Everyone should be empowered to benefit from the advantages of algorithmic and artificial intelligence systems including by making their own, informed choices in the digital environment, while being protected against risks and harm to one's health, safety and fundamental rights.

We commit to:

- a. promoting human-centric, trustworthy and ethical artificial intelligence systems throughout their development, deployment and use, in line with EU values;
- b. ensuring an adequate level of transparency about the use of algorithms and artificial intelligence, and that people are empowered to use them and are informed when interacting with them;
- c. ensuring that algorithmic systems are based on adequate datasets to avoid discrimination and enable human supervision of all outcomes affecting people's safety and fundamental rights;
- d. ensuring that technologies such as artificial intelligence are not used to pre-empt people's choices, for example regarding health, education, employment, and their private life;
- e. providing for safeguards and taking appropriate action, including by promoting trustworthy standards, to ensure that artificial intelligence and digital systems are, at all times, safe and used in full respect for fundamental rights;
- f. taking measures to ensure that research in artificial intelligence respects the highest ethical standards and relevant EU law.

Some pillar concepts in the AIA

- Transparency
- Explicability
- Autonomy of decision
- Dataset provenance
- Know the type algorithm
- Human oversight
- Certification/standards
- Self-assessment/Risk



Legal Design as new black box

EasyChair Terms of Service

The EasyChair Terms of Service have changed as part of EasyChair compliance with the GDPR. The EasyChair Terms of Service have changed as part of EasyChair compliance with the GDPR. To continue using EasyChair you must agree to our new Terms of Service as shown below.

You must agree to our Terms of Service to continue use

You can download these Terms of Service by clicking on "Download". To agree to these terms, tick the box below and click on "Continue". If you disagree with these Terms [click here to log out](#).

☒ I agree to these Terms of Service

Continue

EasyChair Terms of Service

Thank you for choosing EasyChair!

1. TERMS AND CONDITIONS OF SERVICE

1.1 EasyChair Ltd ("we" or "us" or "EasyChair Ltd"), via its online web service EasyChair ("EasyChair" or "EasyChair Web Site") <https://easychair.org> provides services that allow users:

a) to manage document submission, reviewing, publishing, program generation, content management, registration, user management, email management and monitoring, and accounting for conferences, workshops, journals, books, special issues and any other events or publications;

b) to publish papers, articles, preprints, slides, presentations, videos, teaching material, programs of events, calls for papers and volumes and collections thereof;

(the "Service").

WE ASK THAT YOU READ THESE TERMS AND CONDITIONS OF SERVICE (THESE "TERMS") BEFORE USING THE SERVICE. YOUR USE OF THE SERVICE CONSTITUTES YOUR ACCEPTANCE OF THESE TERMS WITHOUT MODIFICATION. IF YOU DO NOT AGREE TO THESE TERMS, DO NOT USE THE SERVICE.

1.2 If you use the Service on behalf of a company, organisation, or other entity, then:

a) "you" includes you and that entity, and

b) you represent and warrant that you are an authorised representative of the entity with the authority to bind the entity to these terms, and that you agree to these terms on the entity's behalf.

1.3 We reserve the right to update the Service at any time at our discretion with or without notice to you. Such updates are designed to improve, enhance and further develop the Service and may take the form of bug fixes, enhanced functions, new modules, or other forms. You agree to provide such updates and permit us to deliver these to you as part of your use of the Service.

1.4 Additional terms may apply to some of our services. For example, if you use our conference registration module, additional terms apply to your use of this module. All of these are referred to herein as the "Additional Terms". Where the Additional Terms and these terms apply, if there is any contradiction between what the Additional Terms say and what these terms say, then the Additional Terms shall take precedence in relation to that element of the Service.

2.1 We grant you a non-transferable, non-exclusive, non-sublicensable, royalty-free worldwide right and license to use the Service. It does not include the right for you (or any third party) to copy, adapt, modify, resell or repurpose any EasyChair Ltd product, service, or data or the EasyChair Web Site. If you are unclear about whether your use is acceptable under these terms, please contact us. The contact information will be available on the contact page of the EasyChair Web Site.

2.2 Your use of the Service does not create a partnership, joint venture or agency relationship or similar relationship between you and us.

2.3 You are expressly prohibited and shall not permit any third party to reproduce, redistribute, duplicate, copy, modify, decompile, disassemble, modify, sell, trade or resell the Service (including the EasyChair Web Site) for any purpose, unless you have been specifically permitted to do so in a separate agreement with EasyChair Ltd.

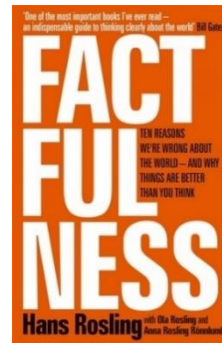
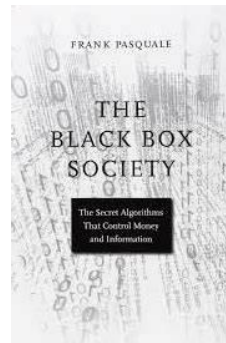
2.4 No competitors or future competitors of EasyChair Ltd are permitted access to the Service and EasyChair Ltd reserves the right to suspend or terminate any account created or used by any person employed by or acting on behalf of any such competitor.

2.5 You must not use the Service to advertise or promote any fictitious conference(s).

3. YOUR USE OF THE SERVICE

3.1 You represent that you are of legal age to form a binding contract and are not prevented from accessing or receiving the Service under any applicable jurisdiction.

3.2 You agree to only use the Service for lawful purposes and as permitted by these Terms.



 **What are your rights in respect of your personal data?**

Your right of data access

 8.1. You are entitled to receive a copy of your personal data that is in our possession (your right of data access).

Your right to erasure and rectification

 8.2 You may request the deletion of personal data or the correction of inaccurate personal data (your right to erasure and rectification). Please note that we may keep certain information concerning you, as required by law, or when we have a legal basis to do so (e.g., our legitimate interest to keep the platform safe and secure for other users).

Your right to object to processing

 8.3 You have the right to object at any time (i) to the processing of your personal data for the purpose of direct marketing, or (ii) to the processing of your personal data for other purposes on grounds relating to your particular situation (your right to object to processing). Please note that in the latter case, this right only applies if the processing of your personal data is based on our legitimate interest.

Your right to restriction to processing

 8.4 You have the right to restrict the processing of your personal data (your right to restriction of processing). Please note that this only applies if (i) you contested the accuracy of your personal data and we are verifying the accuracy of the personal data, (ii) you exercised your right to object and we are still considering, as foreseen by the applicable law, whether our legitimate grounds to process your personal data in that case override your interests, rights and freedoms; or (iii) your personal data has been processed by us in an unlawful way but you either oppose the erasure of the personal data or want us to keep your personal data in order to establish, exercise or defend a legal claim.

Lawyer-readable



Human-readable



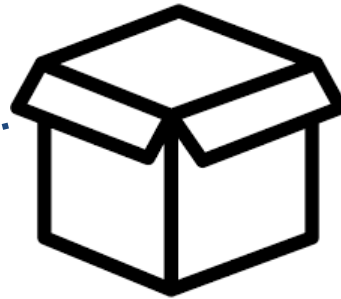
“White box”



AKOMA NTOSO

Architecture for Knowledge-Oriented Management of African Normative Texts using Open Standards and Ontologies

OASIS LegalXML
LegalRuleML



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Lawyer-readable



Machine-readable



Human-readable





Explainable AI/ Legal Design

- The Principle of Explicability: “Operate transparently”

Transparency is key to building and maintaining citizen’s trust in the developers of AI systems and AI systems themselves. Both technological and business model transparency matter from an ethical standpoint. Technological transparency implies that AI systems be auditable,¹⁴ comprehensible and intelligible by human beings at varying levels of comprehension and expertise. Business model transparency means that human beings are knowingly informed of the intention of developers and technology implementers of AI systems.

Explicability¹⁵ is a precondition for achieving informed consent from individuals interacting with AI systems and in order to ensure that the principle of explicability and non-maleficence are achieved the requirement of informed consent should be sought. Explicability also requires accountability measures be put in place. Individuals and groups may request evidence of the baseline parameters and instructions given as inputs for AI decision making (the discovery or prediction sought by an AI system or the factors involved in the discovery or prediction made) by the organisations and developers of an AI system, the technology implementers, or another party in the supply chain.

Explain what ? Explain to whom?

- **What:**
 - Input, process, output, rationale of the decision
- **Whom:**
 - Expert/Judge
 - Common John
 - Minors/vulnerable people/special category
- **How:**
 - Reasoning
 - Convincing
 - Arguing
 - Imposing



Explainability is not easy

- **Explainability** – the status of being explainable
- **Explainable** – that can be explained or understood
- **Explainer** – agent that explains
- **Explaining** – action devoted to tell somebody about something in a way that makes it easy to understand
- **Explanation** – output of the action of explaining that tells why something happened
- **Explicability** – explainable and accountable

- An **explainable** system does not imply **user-friendly explanations**
- Some **explainable** systems are **hard** to understand even **for an expert**

Legal Design: a possible methodology

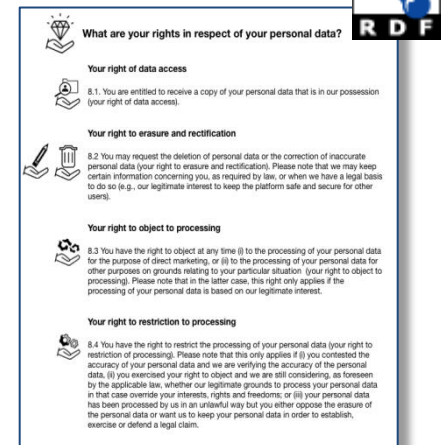
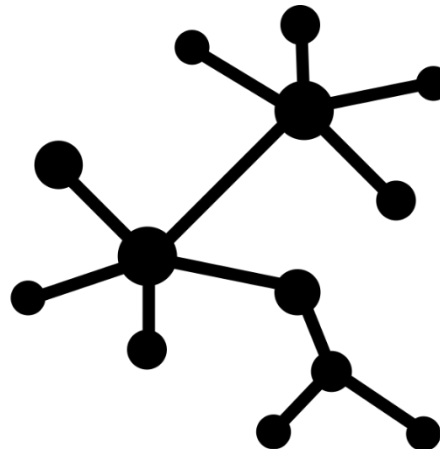
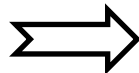
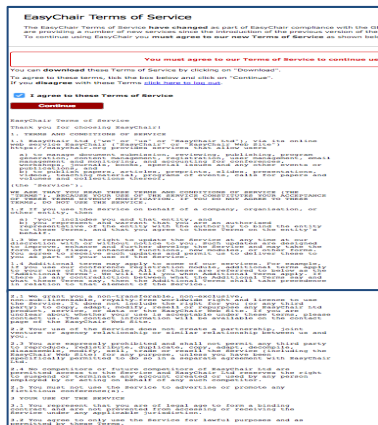
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 - Legal text/message analysis
 - Hypothesis of solution
 - Legal concepts extraction ****
 - » Theory of law check ***
 - » **Legal Ontology analysis *****
 - Simplification of the scenario
 - Usability (**HCI**)
 - Visualization/Solution (**Design**)
 - Communication (**Language**)
 - Testing
 - Interpretation of the results
- Evaluation of the effect on the problem

SEMANTIC



Legal Ontology=bridge between legal text and visual representation

- Legal ontology is a conceptualization method to:
 - decompose complex legal concept
 - discover relationships between legal concepts
 - detect patterns in legal concepts and to simplify
 - check the validity of the solution using the knowledge of robust patterns already done by experts
 - provide machine-readable information that we have lost in the simplification activity (**SEMANTIC**)



Optimized Funding Cycle for Research Articles

Maximum dissemination, economic efficiency and social impact



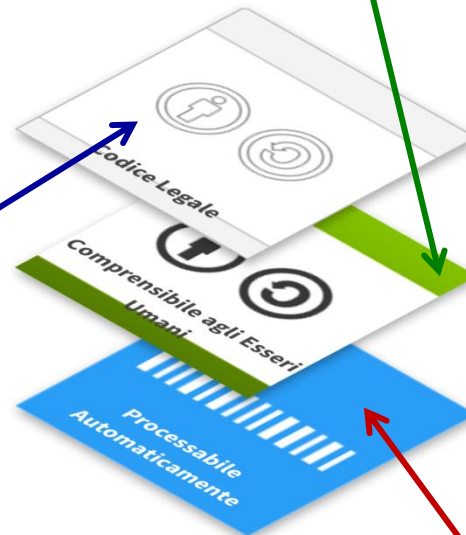
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Visual map in the Web



For the Human we use ICONS

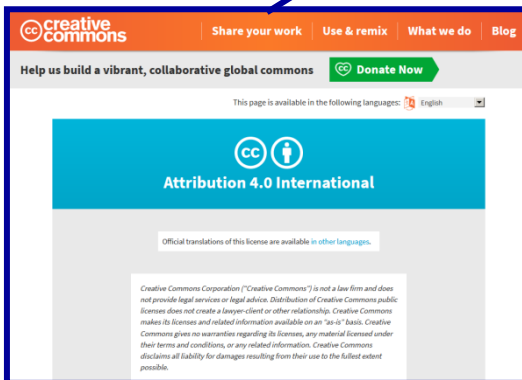
For the Lawyer we use LEGAL TEXT



For the Machine we use CODE

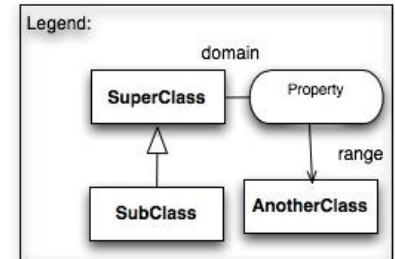
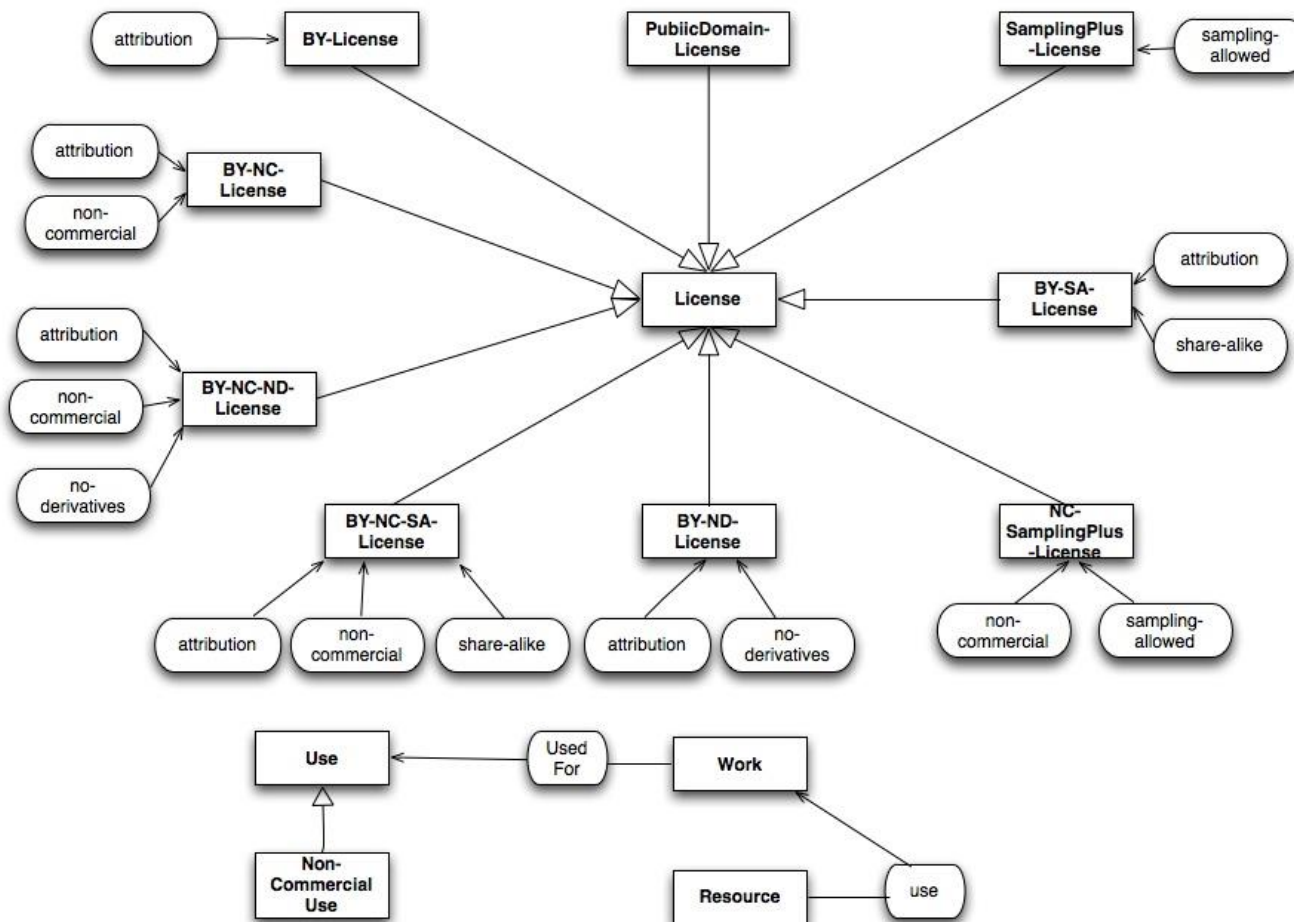
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href="http://creativecommons.org/licenses/by-nc/4.0/">Creative Commons Attribution Non-Commercial 4.0 License</a>.
Permissions beyond the scope of this license may be available at <a
xmlns:cc="http://creativecommons.org/ns#" rel="cc:morePermissions"
href="https://myContent.org">myContent</a>.
    
```



For the machine we use ONTOLOGY

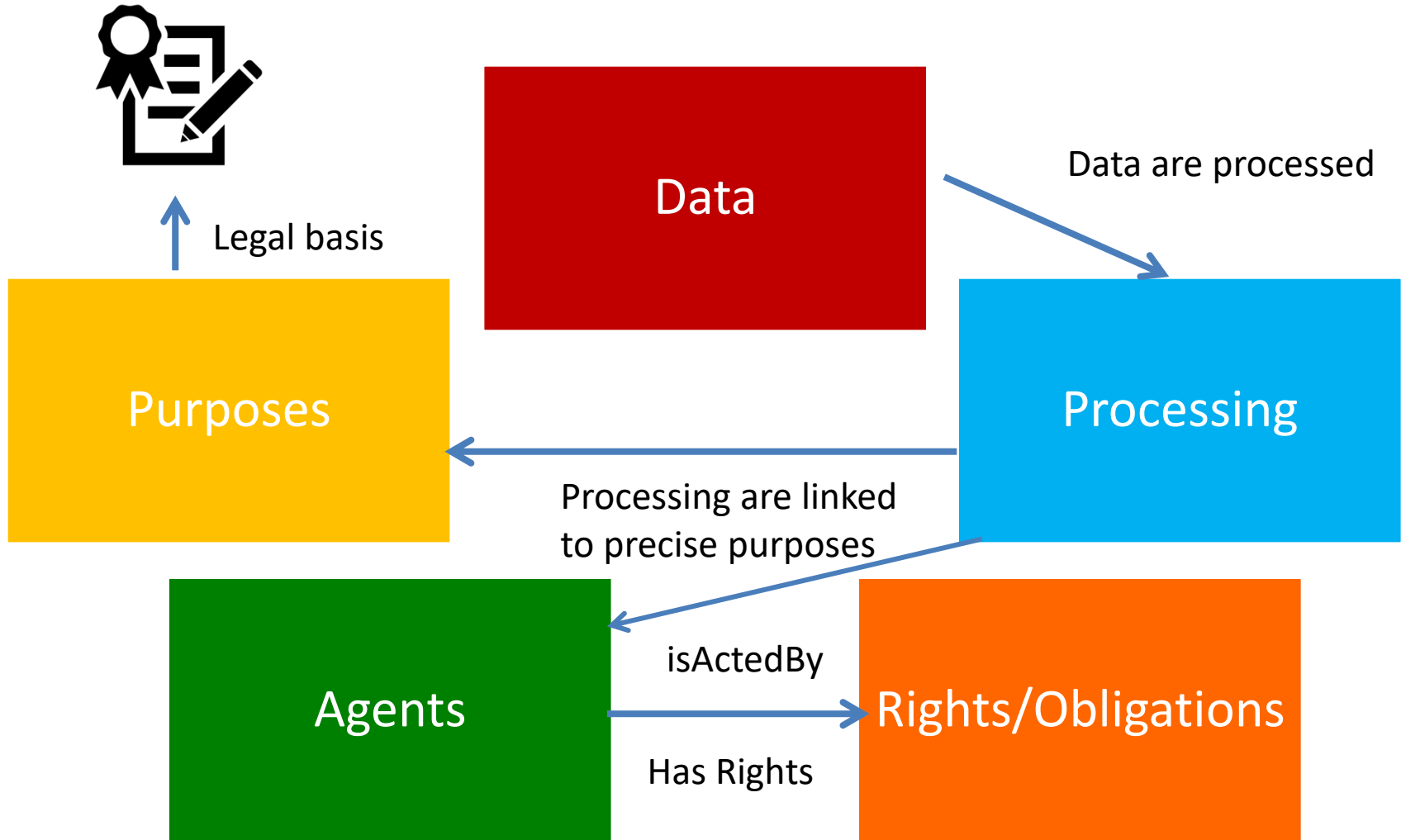
CC RDF schema



EXAMPLE: PrONTO and DaPiS

- **Problem:** provide a better privacy policy comprehension using icons
 - Legal analysis of GPDR
 - Create a legal ontology of the domain
 - Create ICONS accordingly
 - AI and NLP extract the legal concepts from the legal text using the legal ontology (semantic, legal knowledge)
 - Transform the text in a simplified text/visualization/presentation
 - Codify this legal knowledge for the machine in order to provide a correct feedback to the AI

PrOnto ontology of GDPR



attention.png

rightAccess.png

rightClaimAuthor
ity.png

rightObject.png

rightPortability.p
ng

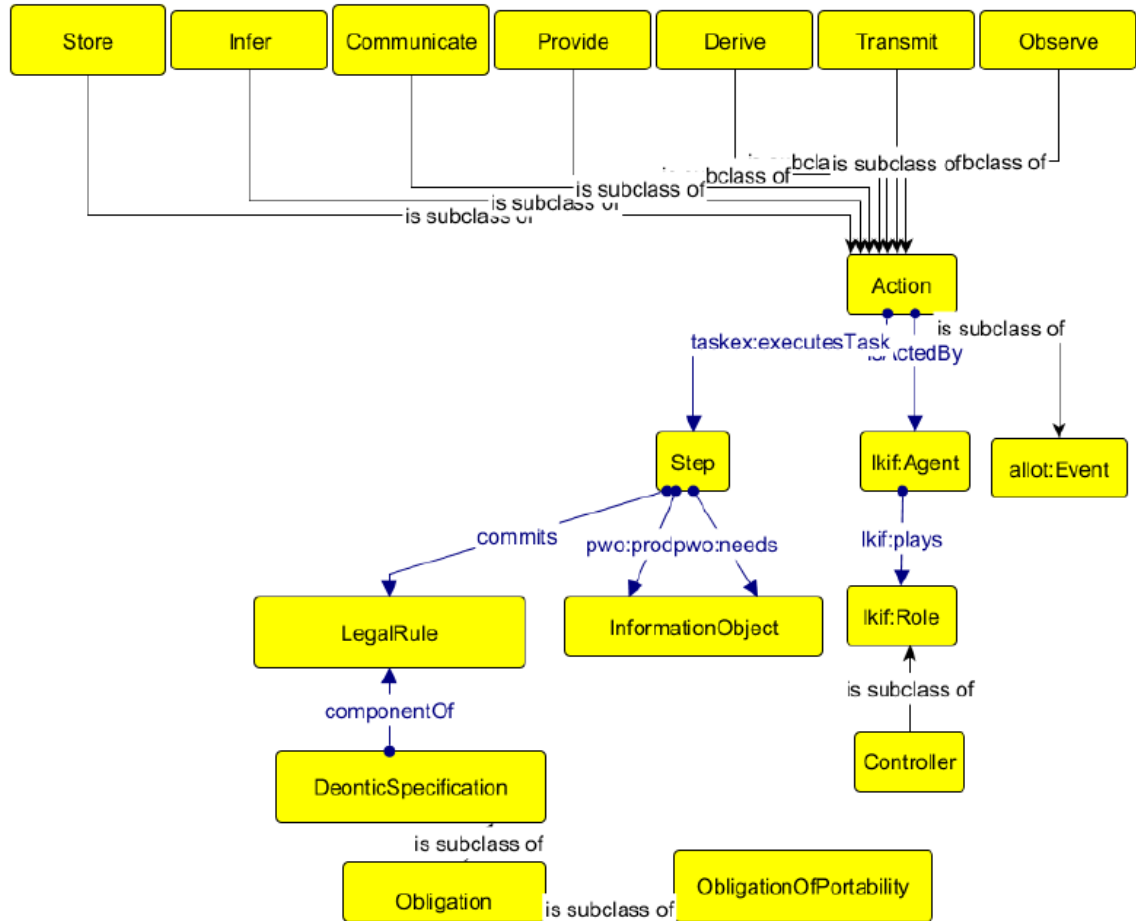
rightRectification.
png

rightRestriction
.png

rightSubject.png

rightToBeInformed.png

rightWithdrawC
nsent.png



Example: privacy policy di Supercell

Informativa Privacy di **SUPERCELL**

Normativa sulla Privacy - Contattaci - I dati che raccogliamo - Perché raccogliamo i dati degli utenti? - Chi può visionare i dati degli utenti - Trasferimenti internazionali di dati - Diritti e opzioni dell'utente - Cookie e tecnologie simili - Come vengono protetti i dati dell'utente - Minori - Per i soggetti che risiedono in California

AKN file Akoma Ntoso

Mostrare annunci personalizzati



Per mostrare annunci pubblicitari personalizzati all'interno del Servizio, nonché in altri siti web e servizi (inclusa la posta elettronica), abbiamo il legittimo interesse a elaborare i dati necessari per:

- monitorare i contenuti a cui accede l'utente in relazione al Servizio e il suo comportamento online;
- consegnare, indirizzare e migliorare la nostra pubblicità e il Servizio;

Per informazioni su come disattivare gli annunci personalizzati, vedere la successiva sezione "Diritti e opzioni dell'utente".

NORMATIVA SULLA PRIVACY





FREEDOM OF CHOICE

People should benefit from a fair online environment, be safe from illegal and harmful content, and be empowered when they interact with new and evolving technologies like artificial intelligence.

#3 LEGAL DESIGN FOR DIGITAL RIGHTS IN AI AND XAI

User persona: citizen

- Definitions of AI system, GPAI, High risk, systemic Risk
- Art. 5 – to inform/explain the citizens about the unacceptable applications of AI system
- Annex III HIGH RISK – to inform/explain
- Actors involved and the chain of responsibility in the AI system – provided, deploy, end-user
- Rights for the citizens – transparency, oversight, audit, to have human explanation, etc.
- AI application for vulnerable categories (minors, women, minority groups, etc.)
- Disclaimer for AI system like chatbot or creation of content (e.g., images, video, etc.)
- XAI art. 13-14 art. 86

INSTRUCTIONS

Goal of Hackathon

- To select one topic
- To select legal regulations connected in AKN
- To develop a little ontology
- To provide prompting engineering
- To produce using LLM legal design informative material

Logistics and Operative issues

- May 30 in person (one member is admitted remotely)
- May 30 23:59 work team with preliminary outcome

Instructions

- The selected teams must send an email to lexai@unibo.it by 11:59 PM on May 30, 2026. The email must contain a link to the software material uploaded to GitHub, including the input data, output data, visualizations, and explanatory documentation for the project and software. The template (available on the hackathon website) must also be attached. The template must include:
 - the names of the participants;
 - the project title;
 - a description of the problem, objectives, methodology, technical solution, and results, all in 5,000 words;
 - the open data license cc-by 4.0 international;
 - a signed authorization for the University of Bologna to view the project and use its contents for the purposes of disseminating the results and awarding the prize.

Sources Online

- [Useful Databases | Center for Legal Data Science | UZH](#)
- <https://github.com/openlegaldata/awesome-legal-data>
- Git data of AKN from UNIBO